

Ejercicio Extra

```

void nroPerfecto(int N) {
    int sumaDivisores, i, j;
    for ( i = 1 ; i < N ; i++) { ← 4OE
        sumaDivisores = 0; ← 1OE
        for ( j = 1 ; j < i ; j++) ← 4OE
            if ( i % j == 0 ) ← 2OE
                sumaDivisores = sumaDivisores + j; ← 2OE
        if ( i == sumaDivisores ) ← 1OE
            System.out.println( "El numero ", i, " es perfecto"); ← 1OE
    }
}

```

$$T(n) = 1 + 1 + \sum_{i=1}^{n-1} (1 + 1 + 1 + \sum_{j=1}^{i-1} (2 + 2 + 1 + 2)) + 1 + 1 + 1 + 2$$

$$T(n) = 2 + \sum_{i=1}^{n-1} (8 + \sum_{j=1}^{i-1} (7))$$

$$T(n) = 2 + \sum_{i=1}^{n-1} (8 + (i - 1 - 1 + 1) * 7)$$

$$T(n) = 2 + \sum_{i=1}^{n-1} (8 + (7i - 7))$$

$$T(n) = 2 + \sum_{i=1}^{n-1} (8 + 7i - 7)$$

$$T(n) = 2 + \sum_{i=1}^{n-1} (7i + 1)$$

$$T(n) = 2 + \sum_{i=1}^{n-1} (7i) + \sum_{i=1}^{n-1} (1)$$

$$T(n) = 2 + 7 * \sum_{i=1}^{n-1} (i) + \sum_{i=1}^{n-1} (1)$$

$$T(n) = 2 + 7 * \left(\frac{(n-1-1+1)*(n-1+1)}{2} \right) + (n-1-1+1) * 1$$

$$T(n) = 2 + 7 * \frac{n^2-n}{2} + n - 1$$

$$T(n) = 2 + 7 * \frac{7n^2}{2} - \frac{7n}{2} + n - 1$$

$$T(n) = \frac{7}{2}n^2 - \frac{5}{2}n + 1 - \text{COMPLEJIDAD CUADRÁTICA}$$

Grafica en la siguiente página:

